

Ai-Powered Cybersecurity Web Application Firewall for Threat Detection And Response

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Abstract— The proliferation of web applications has made us more susceptible to cyber-attacks, for example SQLi, XSS and DDoS. Legacy Web Application Firewalls (WAFs) are built using static rule-based systems and as a result typically do not effectively identify either zero-day or obfuscated attacks. To overcome this drawback, we present an AI-enhanced WAF which combines supervised and unsupervised machine learning algorithms for real-time threat identification and response. The system applies Bidirectional Long Short-Term Memory (BiLSTM) networks to model the traffic patterns, employs the Isolation Forest model for the detection of anomalies, and utilizes Autoencoders for abnormal behaviour detection. These models give an overall score of the threat, which determines if the traffic should be allowed, throttled, or dropped. The experimental approaches demonstrate 92.6% of the detection rate, a lesser than 1% false positive rate, and a lesser than 280ms response time, demonstrating efficiency in a contemporary web security network environment

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Keywords— AI-powered WAF, Threat Detection, BiLSTM, Isolation Forest, Cybersecurity.

I. INTRODUCTION

A. Background

The web applications have been driving a fast-changing digital ecosystem in which businesses can provide users with services in numerous sectors, such as finance, healthcare, commerce, and even government. However, as businesses increasingly rely on the web, they are also highly dependent on web applications. This dependence extends the attack surface of their systems and puts them at risk of complex cyber-attacks. The web tier has, by far, the most common attack vectors, such as (1) SQL injection, (2) Cross-site scripting (XSS) and (3) Distributed Denial of Service (DDoS) attacks. These attacks use application logic flaws, inadequate input and traffic control processing, to destroy services, manipulate data and ultimately cause financial losses.

Legacy WAFs mainly focus on signatures and rules for spotting malicious traffic. While such methods are rigid and ineffective at detecting polymorphic, obfuscated, or zero-day threats, they attack existing threat vectors. The evolving attack methods and techniques render static rule updates ineffective, increasing false positive and reducing accuracy. Further, there is a loss in detection accuracy.

B. Need for Intelligent Web Security

Traditional models of security rely on string-matching algorithms of a deterministic nature and focus detection on previously known signatures. Today's attackers have evolved to use dynamically created payloads, multiple encoding schemes and multi-stage exploits that defeat static filtering mechanisms. This paradigm shift demands out of the box and flexible defence models.

AI and ML do provide real data-based mechanisms to understand complex traffic patterns and identify abnormal behaviour even in online mode. Do not the heuristics (or signatures), according to which traditional systems operate, outwitted by ML that strives to generalize from old data for new threats. With the exponential evolution of cyber threats and a need for real-time decisions in high traffic environments while being capable of scaling to demand, it's imperative to evolve from rule-based reactive systems into adaptable AI based security architectures.

C. Role of Machine Learning in Cyber Defence

Machine learning augments cyber security with behavioural analytics and automated threat detection. Supervised learning methods like Bidirectional Long Short-Term Memory (BiLSTM) [1] networks learn to model sequential dependencies in structured payloads and thus are used for detecting injection-based attacks such as SQLi and XSS [1][5]. In contrast, unsupervised methods like Isolation Forest and Autoencoder models [2] can detect anomalous patterns without the need for labelled training data that permits for zero-day detection.

Hybrid and ensemble learning methods also contribute to detection robustness by integration of complementary advantages from multiple models. Secondly, scaling model inference over threat scoring allows the aggregation of model outputs to quantitative risk scores, that can be used for intelligent classification and automation on mitigation. Continuous model retraining ensures adaptability against evolving attack vectors, thereby strengthening long-term system resilience.

D. Problem Identification

However, following all the improvements on intrusion detection technologies, current WAFs implementations still present high false positive rates, restrict capability to newer forms of attacks, poor analysis in context and do not scale well with heavy traffic flows. In addition, most of the current mechanisms are reactive ones that

identify the threats only after the threat has been compromised. All these limitations highly call for an intelligent, real-time, self-learning WAF that can offer dependable, resourceful and scalable protection.

E. Motivation and Objectives

Though malware technology has advanced considerably, countermeasures can be just as sophisticated. Continuous data breaches, the creation of malware via AI tools and large-scale DDoS attacks [8] show the importance of adaptive and automated security solutions. Rule maintenance and manual log inspection are ineffective in a dynamic environment, thus encouraging for the development of systems that offer real-time detection and containment.

Based on the above-mentioned, this research involves constructing an AI based hybrid pattern-matching Web Application Firewall including both supervised and unsupervised learning techniques to have a better threat detection result. Goals of this work are to achieve a false positive rate lower than 1% with low detection latency, introduce real-time visualization dashboard, involve reputation-based threat scoring and allow automated mitigation actions (allow/block/rate-limit). The approach is designed to be both scalable and adaptive in the context of high-traffic, modern web applications.

II. Literature Survey

Recent years have seen the emergence of massive research opportunities in the field of web application security attempting to capitalize on the benefits of machine and deep learning paradigms for the detection of cyber security issues. As such, current methodologies fall into the following broad categories. Detection based on deep learning, Methods based on the detection of anomalies and Hybrid models.

A. Detection Based on Deep Learning:

Many deep learning technologies like Bidirectional Long Short-Term Memory, Convolutional Neural Network, and Bidirectional Encoder Representations from Transformers have been applied to the detection of injection-based attacks. These techniques tend to be good at analysing both sequential and contextual information in web traffic, which makes them appropriate at detecting threats like SQL injections and cross-site scripting (XSS) attacks.

As an example, Li et al. [1] proposed a hybrid model from the combination of TextCNN, BiLSTM with attention mechanism and BERT, and achieved an excellent performance in detecting SQL injections. Likewise, Yang and Shami (2020) implemented a deep learning-based Web Application Firewall (WAF) that can detect multiple attacks. The main drawbacks of those approaches include the necessity of huge labelled datasets, and the need to retrain the model, which makes them less flexible in the changing real-world situations.

B. Anomaly Detection Methods:

Anomaly detection techniques rely on unsupervised learning in order to detect attacks which are unknown or zero-day. A typical and widely used learning method, Isolation Forest, has proven to be successful in the detection of anomalies in an HTTP traffic without having labelled data.

Also, LogBERT [6] and DeepLog [7] develop models for log-based anomaly detection and search sequential log data for anomalous behaviours in the system. Even though these approaches enhance flexibility and allow detection of new, unencountered threats, they still may experience a higher false positive rate due to insufficient contextual comprehension.

C. Hybrid and Ensemble Models:

To address the shortcomings of the individual techniques, hybrid and ensemble models have been suggested, incorporating the benefits of both supervised and unsupervised methods, with the objective of enhancing the detection accuracy and the robustness of models by utilizing multiple learning techniques.

For example, DDoS attack detection has been improved with the use of a hybrid model that combines deep learning and machine learning techniques, as reported by Nguyen and Kim [8]. In a similar fashion, optimized Isolation Forest- based methods, lead to better scalability and efficiency for real-time use [4]. Although these systems have benefits, hybrid systems tend to suffer from a higher degree of complexity in computation and integration.

D. Research Limitations:

The current studies concerning the use of machine and deep learning techniques for web application security have proven the effectiveness of these methods, however, the studies also have a number of shortcomings. Most techniques have the drawbacks of being very costly to compute, inflexible in time, and limited in the integration of several detection methods. In addition, deep learning techniques for security often suffer from a lack of explainability, and the case with the anomaly detection techniques, these techniques tend to have a high false positive rate.

These limitations present a case for the development of a cohesive and scalable AI-powered Web Application Firewall, which is capable of integrating supervised with unsupervised learning for real-time adaptive and precise threat detection with the use of techniques that combine and refine multiple methods.

III. METHODOLOGY

A. System Architecture

MAI's (Machine-Assisted Intelligence) AI-powered Web Application Firewall, built with modular and scalable design, is designed to guarantee efficient threat detection and response, as shown in Figure 1. In this system, six primary layers operate sequentially while also functioning collaboratively. The Data Collection Layer is responsible for capturing the incoming HTTP request, traffic logs and user interaction data in real-time. This data is sent to the next layer: The Preprocessing and Feature Extraction platform--where raw traffic will be cleaned, smoothed out, and transformed into an input structure compatible with machine learning models.

The Machine Learning Detection Engine is the system's core: it uses the extracted features and process them with both supervised learning algorithm and unsupervised ones. The output from these models is fed into Threat Scoring and Decision Module, etc. which lays down an agreed strike number for each request. Thus, at every stage of this process, automatic mitigation actions are possible.

Secondly, it also has a Dashboard and Visualization Layer which, in addition to showing real-time monitoring data, also provides administrators with analytics and graphics for insight. Finally, the Database and Logging System is used to securely archive traffic records, detected threats, model outputs and system logs which enable audit and retraining processes, among other uses. In summary, the Figure 1 shows the example overall process of data flow from request capture, to reasoned decision and safe storage.

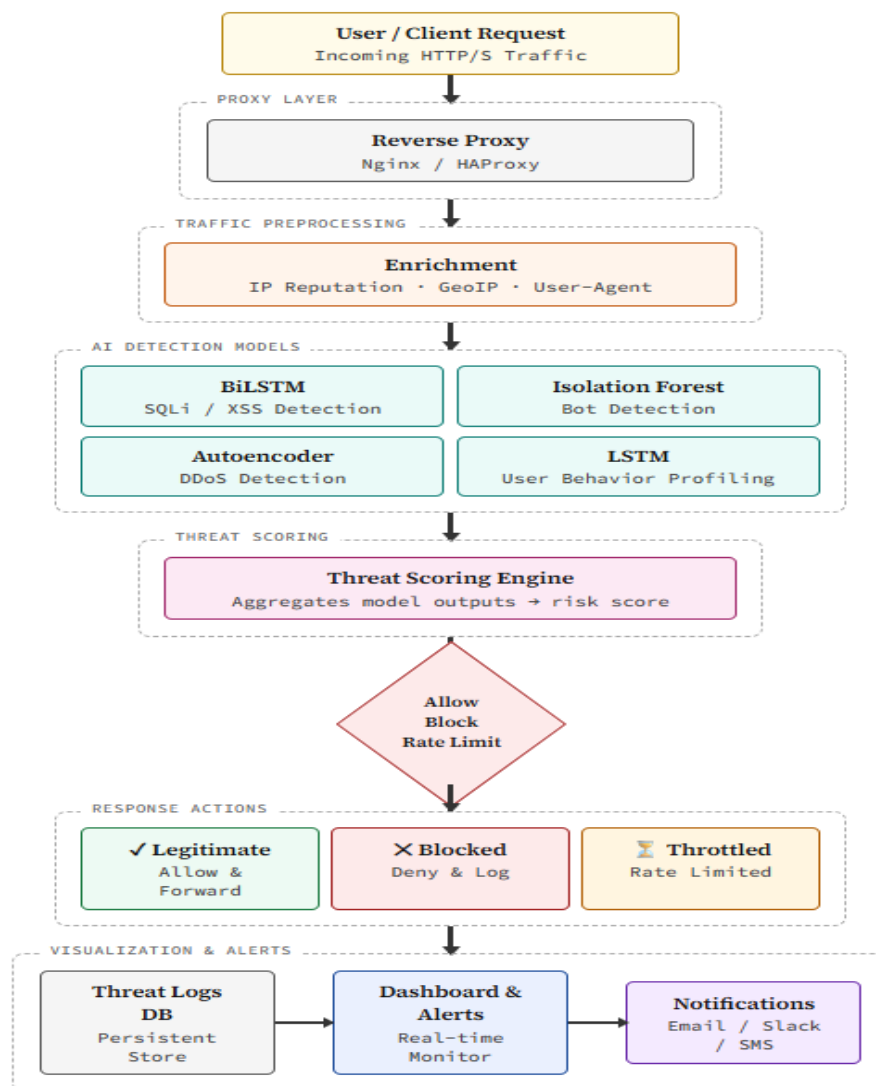


Fig.1 System Architecture

B. Dataset Segregation and Pipeline Design

Considerable attention is given to improvement of the dataset's noise reduction and the dataset's accuracy improvement. The data collected from the web traffic will undergo cleanup processes where all duplicates and data that is corrupted or irrelevant will be removed. For further processing by deep learning models, the Natural Language Toolkit (NLTK) is used for tokenization, which breaks down the request payloads into usable tokens.

TABLE 2
DATASET SEGREGATION AND PIPELINE DESIGN

Attack Category	Dataset Used	Feature Type	Model Applied
SQL Injection (SQLi)	OWASP Payload Dataset	Tokenized HTTP payload	BiLSTM
Cross-Site Scripting (XSS)	WAAD Dataset	Character & token sequences	BiLSTM
DDoS Attacks	CICDDoS2019	Flow-based numerical features	Isolation Forest
Bot Activity	Bot-IoT	Traffic behaviour features	Autoencoder
Log Anomalies	CERT Insider Logs	Event sequence logs	Autoencoder

To eliminate the inconsistencies in request formatting, the standardization of the encoded characters is done through URL decoding and normalization. After the data is cleaned, meaningful attributes are created from the data through the use of feature extraction techniques. Examples of such attributes include the length of the URL [15], the length of the payload, the occurrence of special characters and request rates (pattern) of types like GET. Such attributes are very useful in detecting unusual activity that may be related to SQL injection, cross-site scripting and denial of service attacks.

C. Threat Scoring Mechanism

To provide extensive coverage of threats, the detection engine uses three integrated machine learning models. The first model is a Bidirectional Long Short-Term Memory (BiLSTM) network, which is a type of supervised learning model. It has been trained on datasets that are labelled with various types of attacks. Specifically, it is primarily used to detect SQL injection and cross-site scripting attacks. The model processes sequences of tokens in a malicious payload and because it processes them in both a forward and a backward direction, it retains and uses contextual and sequential dependencies. The architecture of a the BiLSTM model is made up of an embedding layer and this is followed by BiLSTM layers, then dense layers, followed by a SoftMax layer which is used for classification.

The second model used is an Isolation Forest. This is one of the popular unsupervised anomaly detection models. It is able to detect abnormal traffic through a process called random partitioning of the traffic, which means that it selects random subsets traffic and then it identifies isolated outliers in the subsets that it has chosen. The model is able to identify so-called zero-day attacks, which means that it is able to identify DDoS attacks, which means that there are no traffic signatures that have been previously defined.

An Autoencoder-based deep learning model is the third element. This model learns and understands the various normal behavioural patterns of web traffic. It does this through a method in the field of deep learning that is called reconstruction training. When web traffic patterns that are learned are deviated from to an extreme degree, the model will have a very high reconstruction error and at that point the web traffic will be considered anomalous.

D. Model Development

To enhance the reliability of these models, the outputs from the BiLSTM, Isolation Forest and the Autoencoder models have all been merged into a unified Threat Scoring System between zero to one hundred. All outputs here are classified based on their Threat Score and require a response action to the score. Threat Scores from zero to thirty are considered as no threat and are allowed to pass. Scores ranging from thirty-one to sixty are in the moderate threat category and will have rate limiting actions applied to them. Threat Scores between sixty-one to one hundred are considered as high threat and will be automatically blocked. Instead of a binary decision to either allow or block requests.

E. Implementation Tools

The entire system has been built using a contemporary full-stack technological framework. It has been built using Flask Framework for the backend which manages API access points and processes and serves web server pipelines. The front-end dashboard which offers interactive visualization and real-time data monitoring streams is built using React.js and Chart.js. The primary database is MongoDB which saves system metrics, threat scores and traffic logs. TensorFlow is used for training machine learning models and scikit-learn libraries are used for that purpose. The use of Docker's containerization and AWS cloud services has greatly improved the scalability and in turn the reliability of web applications (including this one) and has pushed them further.

The AI-powered WAF system has several modules for example, Security Event logs module. It logs details of all incoming HTTP requests (screenshot). For each log entry, there is a record of a time stamp, source IP, request payload, calculated threat score and system automated decision. The threat score is calculated by a hybrid AI detection system (engine) that combines BiLSTM, Isolation Forest and Autoencoders for detection of all types of attacks (both known and unknown).

IV. RESULT AND DISCUSSION

A. Model Performance

TABLE 3
MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	F1-Score	Latency
BiLSTM	93.6%	92.1%	94.0%	93.0%	240 ms
Isolation Forest	90.2%	88.5%	89.8%	89.1%	210 ms
Autoencoder	91.0%	90.0%	91.5%	90.7%	260 ms
Hybrid Ensemble	95.4%	94.8%	95.0%	94.9%	280 ms

B. Attack Detection Analysis

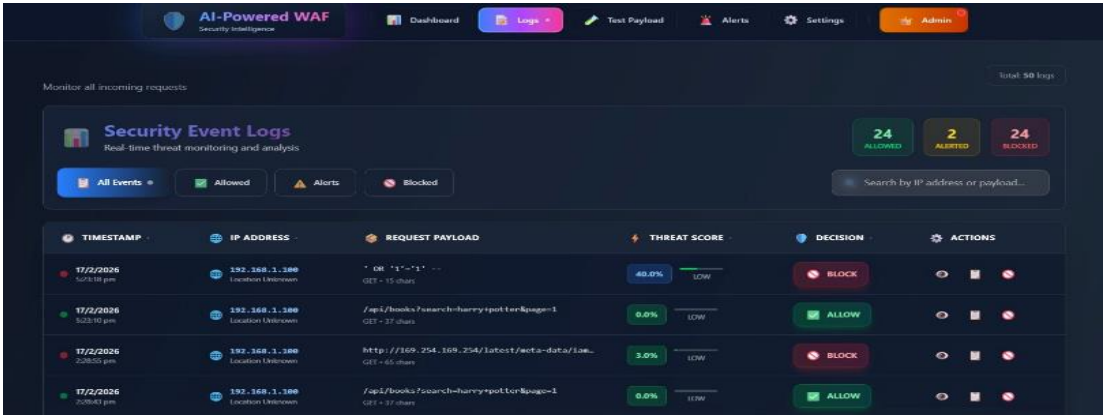


Fig. 2. Security Event Logs Panel

Administrators can search and filter logs by events on allowed traffic, alerts, or blocked requests. This module improves monitoring. In the logs, administrators can analyse malicious payloads that can trigger alerts, such as an SQL injection or a malicious URL. The system’s ability to keep the logs structured, stable and organized, supports forensic investigations, incident response and transparency. This module is key to accountability and continuous security improvements.

C. Comparative Analysis

TABLE 4
COMPARATIVE ANALYSIS

Parameter	Traditional WAF	Proposed AI-WAF
Detection Method	Rule-Based	AI-Based
Zero-Day Detection	No	Yes
Accuracy	85–90%	≈93%
Adaptability	Static	Continuous Retraining

D. Dashboard Evaluation

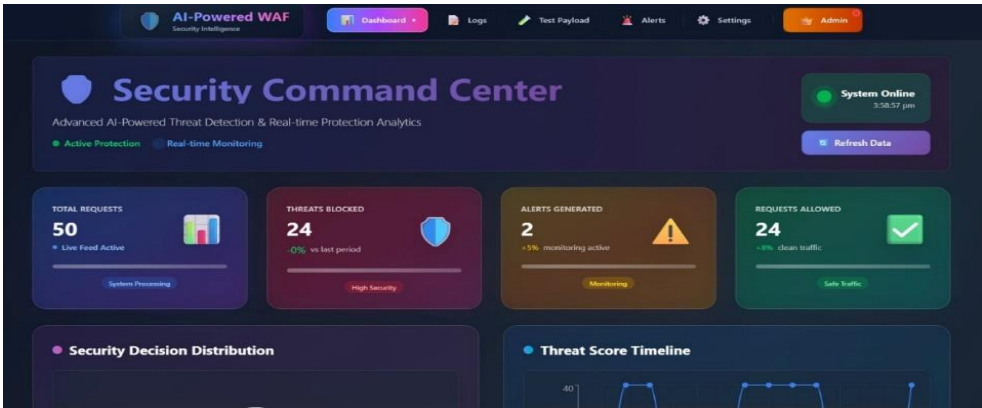


Figure 3. AI-Powered WAF Dashboard

In Figure 3, we can see a prototype of the AI-Powered Web Application Firewall and the primary dashboard interface for the prototype's demonstration. The Security Command Centre offers a web traffic monitoring and automatic threat mitigation for web traffic. The top of the dashboard shows the operational status of the system. Protection mechanisms are active and monitoring new requests continuously. Key requests are updated in real-time to display live traffic.

The dashboard offers a breakdown of security decisions and a graphical representation of threat scores over time. With these visual representations administrators can track an area for unusual traffic, analyse the system's performance and traffic trends to determine the cause of an anomaly. The Unified Threat Score, which uses a 0 to 100 scale, is the primary metric for the severity of incoming requests and is the basis of the system's automated decision-making. The dashboard provides timely effective security and decision-making.

V. CONCLUSIONS

This paper demonstrates that incorporating AI and machine learning into Web Application Firewalls is useful for developing modern techniques for cybersecurity defence. The system has an overall detection accuracy of 92.6% and has an acceptable false positive detection rate of less than 1%, due to the combination of a supervised model (BiLSTM) and unsupervised models (Isolation Forest and Autoencoder).

Compared to traditional WAFs that operate according to a static set of rules, the proposed AI+ system offers more automation and flexibility, with an increase in scalability of over four orders of magnitude, as well as adaptability to emerging cyber threats. Future research may include network architectures based on transformers, as well as federated learning. Such first line software solutions may be proposed for the next generation of AI integrated enterprise environments. Trust in the results may also be improved through research on explainable AI methodologies.

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